

Intro to Physics!

SciTeens SOLID Physics Day 1





Meet the Team

Co-Directors:

Aarti Kalamangalam - She will be attending University of Florida this Fall where she's planning to study Linguistics and Computer Science. Aarti chose this major because of her love for problems solving, words and learning new languages!

Ohm Parikh - He plans to begin studying Biological Sciences this fall as a freshman in college. Ohm chose this major because he loved biology in high school and hopes to work on biomedical research in the future!

Teaching Assistant:

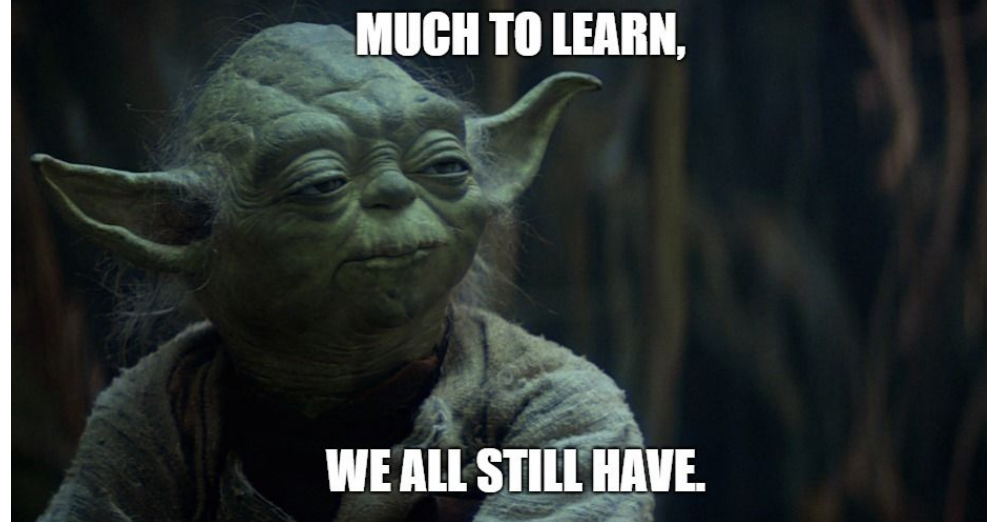
Ashley Pelton - She plans to study Neuroscience this fall as a freshmen at Georgia Tech. Ashley chose this major because she has always loved neuroscience and hopes to be a neurologist in the future!





Learning Objectives

- Vectors
 - Vectors vs Scalars, defining both
 - Examples of vectors
 - The vectors of particle motion
- Manipulating Vectors
 - Vector components
 - Operations with vectors





What is Physics?

- Physics deals with the structure of matter and how matter behaves through space and time
 - The scope of the matter can range from atoms and their subcomponents to the entire universe as an object
- Within kinematics (which we will study), many phenomenon can be observed:
 - The motions of cars in drag racing (1 dimensional) using 0-60 or 0-100 values
 - The flight paths of various sports balls based on their angle of elevation when launched (2D)
 - Optimizing route from point A to B based on speed limits and lengths of various roads
 - The flight paths of space vehicles (3D kinematics) and the optimization of travel from Earth to another location



What is Physics? (cont'd)

There are many careers in physics, as well as many areas of study within physics

Areas of study:

- kinematics/particle motion
- thermodynamics
- fluid mechanics
- electricity and magnetism
- quantum mechanics,
- optics and acoustics

Careers involving physics:

- engineering (mechanical, aerospace, nuclear, etc.)
- Physicist
- physics researcher
- physics educator
- astronaut
- quantum programmer



Course of Study

In this course, we will be focusing exclusively on kinematics, which focus on the motion of objects without consideration to the forces causing the motion

We will cover the following physics topics, building from the basics taught earlier to the more complex ideas studied later in this course:

- Vectors, vector components, and vector manipulation
- The vectors in kinematics
- The kinematics equations, when to use each, variables involved in each
- 2 dimensional kinematics, combining vectors and the kinematic equations



The Final Project

The final project will be a culmination of the physics and programming learned during the course. Students will complete the following tasks:

- Create a function plotting the path of a real life particle (baseball, car, spaceship, etc.) over a reasonable interval of time
 - The function should have two distinct parts: one that performs the calculations relevant to generating the positions, velocity, and acceleration arrays (using the kinematic equations), and another that performs the plotting
- Generate plots of the particle's positions and velocity over time using the function
- Analyze the path of the particle using the plots generated and the data gathered, which should be compiled into a dataframe

What are Vectors?

SCALAR AND VECTOR

BYJU'S
The Learning App



SCALAR

A scalar is a quantity that is fully described by a magnitude only. It is described by just a single number. Some examples of scalar quantities include speed, volume, mass and time.



VECTOR

A vector is a quantity that has both a magnitude and a direction. Vector quantities are important in the study of motion. Some examples of vector quantities include force, velocity, acceleration and momentum.



Vectors in Kinematics

The following are the vectors used in kinematics, along with the symbols used to reference them as well as their practical meaning:

- Position (x or y): The displacement of the particle from a predetermined reference point (units: m, km, miles)
- Velocity (v): The rate of change of the position of a particle with respect to time (units: m/s, km/h, mph)
- Acceleration (a): The rate of change of the velocity of a particle with respect to time (units: m/s/s, km/h/h, mph/h)



Checkpoint 1

Of the following, which quantity is a scalar (the rest are vectors)?

- (a) velocity
- (b) speed
- (c) position
- (d) acceleration



Quick Algebra Review

Before we begin manipulating vectors, we should review a few important algebra topics!

<i>Operation</i>	<i>Inverse Operation</i>
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+

-

-

+

$\times$

$\div$

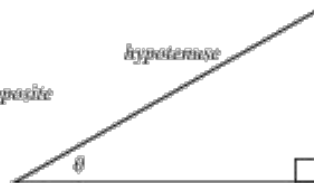
$\div$

$\times$



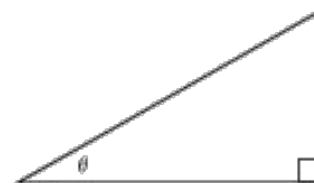
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

SOH



$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

CAH



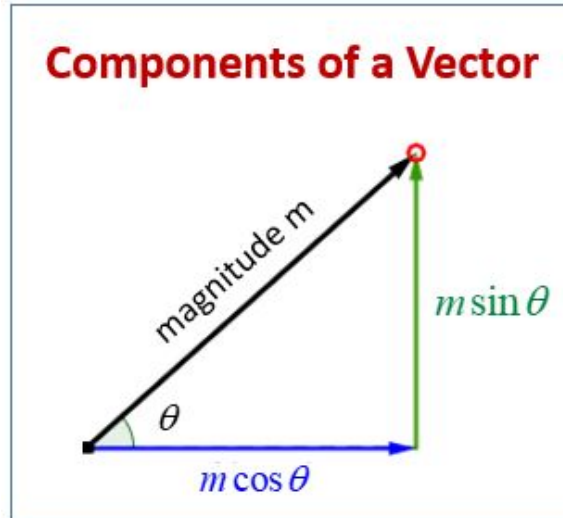
$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

TOA



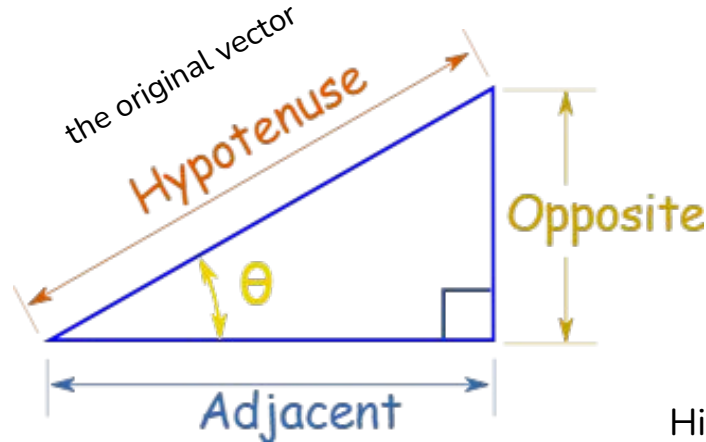
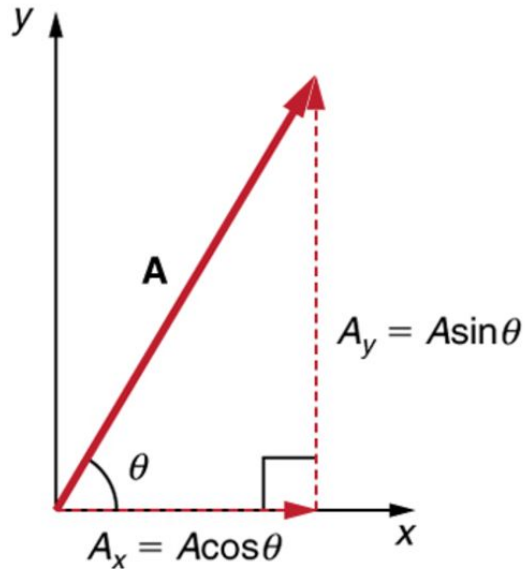
Manipulating Vectors

- In physics, motion is handled in 1 dimension at a time, so values dealing with 2 or more dimensions (ex: 15 m displacement at a 45° angle) must be broken down into their dimensional components (x and y)



Breaking Down Vectors

- On the previous slide, a general example of vector breakdown was given
- Now let's go more into detail and learn how vectors work!



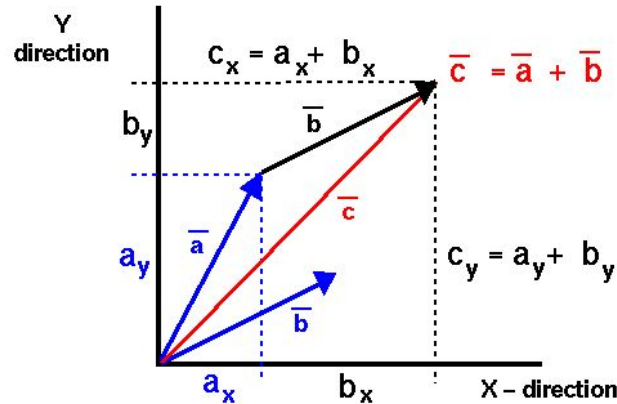
the "y" vector component

the "x" vector component

Hint: Remember
SohCahToa!

Advantages of Vector Components

- As aforementioned, vector components are necessary when studying projectile motion, especially in real life scenarios when 2 or 3 dimensions are involved
- Vector components also make combining vectors much easier:

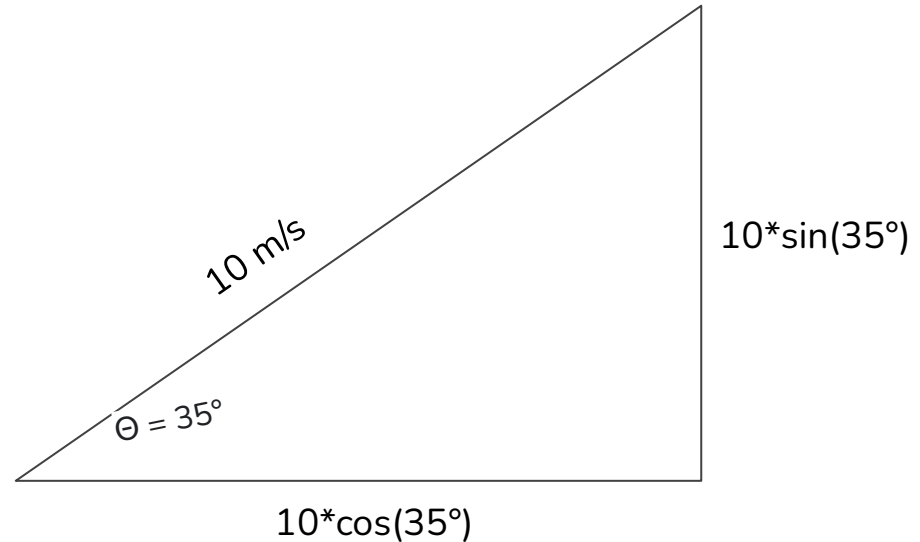




Checkpoint 2

Which of the following gives the correct x and y components of a velocity vector with magnitude 10 m/s and an angle of elevation of $\theta = 35^\circ$?

- (a) x: 10 m/s | y: 10 m/s
- (b) x: 6.66 m/s | y: 3.33 m/s
- (c) x: 5.74 m/s | y: 8.19 m/s
- (d) x: 6.5 m/s | y: 3.5 m/s





Relationships Between Vectors

Given the aforementioned definitions of position, velocity, and acceleration, we can define some simple relationships:

$$(1) \quad a = v/t \quad \text{given } v = 0 \text{ at } t = 0$$

$$(1) \quad v = x/t \quad \text{given } x = 0 \text{ at } t = 0$$

$$(1) \quad a = (x/t)/t \quad \text{given } v = 0 \text{ and } x = 0 \text{ at } t = 0$$



Checkpoint 3: Example Problem

Suppose the position, velocity, and acceleration of a particle are defined as x , v , and a respectively, and position is defined in meters (m) and time is measured in seconds (s)

- (1) If the particle travels 100m in 5s, assuming $a = 0$, what is the initial velocity and the final velocity?
- (2) If the particle travels 100m in 5s, assuming $x = 0$ and $v = 0$ at $t = 0$, what is the acceleration of the particle?



Key Points

- Vectors vs Scalars: Both have magnitude, a vector has magnitude and direction, and it is important to specify both
- The 3 most important vectors in kinematics: acceleration (a), velocity (v), and position (x or y)
- Vector components: Use SohCahToa to generate a right triangle with the original vector as the hypotenuse and the legs as the components
- Operations using vector components: Addition/subtraction of vectors are much simpler with comparable components (ex: x and y components for each vector can be combined)
- Basic relationships between vectors a , v , $x(y)$, and t (time): $v = x/t$, $a = v/t = x/t/t$



Additional Practice/Resources

[Resource #1](#)

[Resource #2](#)

[Resource #3](#)

[Additional Practice #1](#)

[Additional Practice #2](#)